ERGODIC COMPONENTS AND CLUSTERING FOR REFLECTION-POSITIVE FIELDS

Abstract. A translation-invariant law of finite-component tempered fields in three dimensions is assumed invariant and reflection positive at the coordinate and diagonal planes. We prove clustering relative to its translation-invariant sigma-algebra. Almost every translation-ergodic component retains the reflection properties and is mixing in every direction. The proof uses a diagonal reflection to obtain a spectral cone, then identifies zero-energy vectors with translation-invariant observables. If the parent law also satisfies an L^2 Gaussian bound, the same constant holds in almost every component.

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1. Introduction

Let V be a finite-dimensional real inner-product space with an orthogonal representation $\rho : O(3) \to O(V)$. Let $\mathcal{X} = \mathcal{S}'(\mathbb{R}^3; V)$, equipped with the sigma-algebra generated by evaluations. For $g(x) = Qx + b$, use

$$(g \cdot f)(x) = \rho(Q) f(Q^{-1}(x - b)), \qquad (g \cdot F)(f) = F(g^{-1} \cdot f), \qquad \widehat{g}A(F) = A(g^{-1} \cdot F). \tag{1}$$

Write $U(a) = \widehat{\tau_a}$. For open $O \subset \mathbb{R}^3$, let Σ_O be generated by $F(f)$ with real $f \in C_c^\infty(O; V)$. For a reflection r, put $\mathcal{R}_r = \widehat{r}$. An r-invariant law λ is reflection positive on an open side H^+ if

$$\mathbb{E}_\lambda[\overline{\mathcal{R}_r A}\, A] \geq 0$$

for finite character sums $A = \sum_b z_b e^{iF(f_b)}$ with real $f_b \in C_c^\infty(H^+; V)$. The positive sides below are chosen by making the displayed linear coordinate function positive.

Theorem A. Let λ be a translation-invariant probability law on $\mathcal{X}$, invariant and reflection positive at the nine planes

$$x_j = 0 \quad (1 \leq j \leq 3), \qquad x_j - x_k = 0, \quad x_j + x_k = 0 \quad (1 \leq j < k \leq 3). \tag{2}$$

Let $\mathcal{I}$ be the translation-invariant sigma-algebra, with invariance understood modulo λ-null sets, and let $P = \mathbb{E}_\lambda[\,\cdot \mid \mathcal{I}]$. Then

$$\langle A, U(a)B\rangle_{L^2(\lambda)} \longrightarrow \langle A, PB\rangle_{L^2(\lambda)} \qquad (|a| \longrightarrow \infty) \tag{3}$$

for every $A, B \in L^2(\lambda)$.

There is a unique translation-ergodic decomposition $\lambda = \int \lambda_\xi \, d\nu(\xi)$, understood as the distribution of its conditional law on $\mathcal{X}$. On one common ν-full set, every λ_ξ is translation ergodic, invariant and reflection positive at every real translate of the planes in (2), and

$$\langle A, U(a)B\rangle_{L^2(\lambda_\xi)} \longrightarrow \overline{\mathbb{E}_{\lambda_\xi} A}\, \mathbb{E}_{\lambda_\xi} B \qquad (|a| \longrightarrow \infty) \tag{4}$$

for every $A, B \in L^2(\lambda_\xi)$. Reflection positivity holds on the entire positive-side $L^2(\lambda_\xi)$ space. These component laws are cubically invariant.

If in addition, for some fixed $C \geq 0$,

$$\mathbb{E}_\lambda e^{F(f)} \leq e^{C\|f\|_2^2/2} \qquad (f \in \mathcal{S}(\mathbb{R}^3; V) \text{ real}), \tag{5}$$

then the common full-measure set above may be chosen so that every component in it satisfies (5) with the same C, simultaneously for all real tests. Any specified measurable event of full λ-measure has full measure in almost every component. ■□

The proof follows the reflection and component arguments of [Cur, Section 7]. They depend on the law and its geometric action, so the proof applies to the finite component space V above. No field equation or Gaussian estimate is used for (3) or the reflection properties of the components.

The spectral step is essential for the all-direction limit. Coordinate reflection positivity constructs a nonnegative generator H_j and commuting transverse momenta P_i. A second reconstruction at a diagonal mirror gives a bound on Laplace transforms of their joint spectral measure. We use transforms whose suprema on the positive real and imaginary axes have bounded product, while their modulus grows exponentially outside the sector $E \geq |p_i|$. This proves $H_j \geq |\mathsf{P}_i|$. At zero energy all transverse momenta therefore vanish. It follows that invariance under one coordinate translation group is already invariance under all translations. Semigroup convergence then gives (3), uniformly in transverse displacements.

For the component assertions, a bounded translation-invariant observable has a representative in either half-space and is fixed by its reflection. Applying reflection positivity to $\sqrt{h}A$, with h a nonnegative invariant weight, yields a conditional reflection inequality. Countable determining character algebras put the resulting properties on one set of component laws. The optional Gaussian assertion uses a separate estimate: spatial averages of $e^{F(f)}$ have moments controlled by translated L^2 overlaps of f. Those overlaps vanish at large separation, which preserves the original constant C after conditioning.

The conclusion is qualitative clustering. It asserts no positive spectral gap. The hypotheses also do not assert invariance under arbitrary rotations or field-sign change in each component.

2. Reflection quotients

For an open set O, let Σ_O be generated by $F(f)$ with real $f \in C_c^\infty(O; V)$. Let λ be a translation-invariant law and let r_j reflect in $x_j = 0$. Suppose that λ is invariant and reflection positive at this plane. For a Euclidean motion g write $\widehat{g}A(F) = A(g^{-1}F)$ and $U(a) = \widehat{\tau}_a$. For a reflection r put $\mathcal{R}_r = \widehat{r}$. Write $\mathcal{R}_j = \mathcal{R}_{r_j}$ and $\Theta_j A = \overline{\mathcal{R}_j A}$. On $\mathcal{E}_j^+ = L^2(\Sigma_{\{x_j>0\}}, \lambda)$ the form

$$b_j(A, B) = \mathbb{E}_\lambda[(\Theta_j A)B]$$

is positive semidefinite and $b_j(A, A) \leq \|A\|^2_{L^2(\lambda)}$. Let $\mathcal{H}_j$ be its quotient completion, $[A]_j$ the quotient map, and $\Omega_j = [1]_j$. Reflection positivity stated on the characters $e^{iF(f)}$, with real $f \in C_c^\infty(\{x_j > 0\}; V)$, extends to this form: their linear span is dense in the positive-side L^2 space, and the form is bounded by the L^2 norm. The character-density argument is given in section 5.

Proposition 2.1. There are a non-negative self-adjoint H_j and a strongly continuous unitary representation V_j of $e_j^\perp$ on $\mathcal{H}_j$ such that

$$e^{-tH_j} V_j(y)[A]_j = [U(te_j + y)A]_j, \qquad t \geq 0, \quad y \perp e_j. \tag{6}$$

The two representations commute, and $H_j\Omega_j = 0$. ∎

Proof. Translations are strongly continuous on $L^2(\lambda)$. On characters this follows from continuity of translation of a Schwartz test and bounded convergence; their density gives it on all of $L^2(\lambda)$. Positive time translations preserve the positive-side algebra. With $|A|_b = b_j(A,A)^{1/2}$, reflection and translation invariance give

$$|U(te_j)A|_b^2 = b_j(A, U(2te_j)A) \le |A|_b\, |U(2te_j)A|_b.$$

Iteration yields

$$|U(te_j)A|_b \le |A|_b^{1-2^{-k}} \|A\|_{L^2(\lambda)}^{2^{-k}}.$$

Letting $k \to \infty$ proves that $[A]_j \mapsto [U(te_j)A]_j$ is a well-defined contraction. It is symmetric, obeys the semigroup law, and is strongly continuous by the contraction of the quotient map. Writing it as $T_j(t)$, one has $T_j(t) = T_j(t/2)^2 \ge 0$. If $T_j(t)\psi = 0$, repeated square roots and strong continuity give $\psi = 0$. Spectral calculus therefore defines $H_j = -\log T_j(1) \ge 0$; uniqueness of positive roots gives $T_j(t) = e^{-tH_j}$ first at dyadic times and then at all times.

For $y \perp e_j$, $U(y)$ preserves the half-space and commutes with reflection, so it is an isometry for b_j, with inverse $U(-y)$. Its quotient extension is the unitary $V_j(y)$. Strong continuity and commutation with $T_j(t)$ follow from the corresponding Euclidean translation identities. The constant observable is fixed throughout. □

Write $V_j(se_i) = e^{is\mathsf{P}_i}$ for $i \ne j$. The joint spectral measure E_j of H_j and these commuting momenta lies on $[0,\infty) \times e_j^{\perp}$. For $\psi \in \mathcal{H}_j$ let $\varrho_\psi(A) = \langle \psi, \mathsf{E}_j(A)\psi\rangle$. Then

$$\langle \psi, e^{-tH_j} V_j(y)\psi\rangle = \int e^{-tE + ip\cdot y}\, \mathrm{d}\varrho_\psi(E,p). \tag{7}$$

3. The spectral cone from a diagonal reflection

Proposition 3.1. Let $i \ne j$, and suppose in addition that λ is invariant and reflection positive at $x_j + x_i = 0$. Then

$$\mathsf{E}_j\{(E,p) : E < |p_i|\} = 0. \tag{8}$$

Thus $H_j \ge |\mathsf{P}_i|$ in the joint spectral calculus. ∎

Proof. Fix $D > 0$ and $A \in L^2(\Sigma_B, \lambda)$, where $B = \{0 < x_j < D,\ |x_i| < D\}$, and set $\psi = [A]_j$. Let ϱ be the image of ϱ_ψ under $(E,p) \mapsto (E, p_i)$. This is a finite positive measure on the right half-plane, identified by $z = E - ip_i$. For $a, c \ge 0$ consider

$$K(a,c) = \mathbb{E}_\lambda\big[(\Theta_j A)U(2De_j + a(e_j + e_i) + c(e_j - e_i))A\big].$$

The axis reconstruction gives

$$K(a,c) = \int e^{-2DE} e^{-az - c\bar z}\, \mathrm{d}\varrho(E, p_i). \tag{9}$$

There is a second representation with the diagonal as time direction. Put $n_\pm = (e_j \pm e_i)/\sqrt{2}$, $y = D(e_j + e_i)/2$, and let r_+ reflect in $x_j + x_i = 0$. The observables

$$A_1 = \mathcal{R}_{r_+} U(-y)\mathcal{R}_j A, \qquad A_2 = U(2De_j - y)A$$

are supported on the positive side of this diagonal. Indeed, $r_j B - y$ lies in its negative side and $B + 2De_j - y$ in its positive side. Their L^2 norms equal $\|A\|_2$. Writing $\Theta_+ A = \overline{\mathcal{R}_{r_+} A}$, translation of the original expectation by $-y$ gives

$$K(a,c) = \mathbb{E}_\lambda\big[(\Theta_+ A_1)U(a(e_j + e_i) + c(e_j - e_i))A_2\big].$$

Choose $Q \in O(3)$ carrying e_j, e_i to n_+, n_- and apply proposition 2.1 to the transformed law $(Q^{-1})_*\lambda$. If E' is its joint spectral measure, set $\psi_a = [\widehat{Q^{-1}}A_a]'$, $a = 1, 2$, and let ϱ' be the image of $\langle \psi_1, \mathsf{E}'(\cdot)\psi_2\rangle$ under $(E, p) \mapsto (\sqrt{2}E, \sqrt{2}p_i)$. This gives

$$K(a, c) = \int e^{-aE'+icp'} \, \mathrm{d}\varrho'(E', p'), \qquad \|\varrho'\|_{\mathrm{TV}} \le \|A\|_2^2, \tag{10}$$

where $E' \ge 0$. The variation bound is spectral Cauchy–Schwarz applied to the two quotient vectors of A_1, A_2. The factors $\sqrt{2}$ from the diagonal coordinates have been absorbed into E', p'.

For $h \in L^1(0, \infty)$ write $\widetilde{h}(z) = \int_0^\infty h(a)e^{-az} \, \mathrm{d}a$. Integrating the two representations against $h(a)\overline{h(c)}$ yields the decisive bound

$$\int e^{-2DE}|\widetilde{h}(E - ip_i)|^2 \, \mathrm{d}\varrho \le \|A\|_2^2 \sup_{u \ge 0} |\widetilde{h}(u)| \sup_{v \in \mathbb{R}} |\widetilde{h}(iv)|. \tag{11}$$

All integrations are justified by the finite variation of the spectral measures and the L^1 assumption on h.

We choose transforms whose two axis suprema have bounded product, but whose modulus grows exponentially outside $|\arg z| \le \pi/4$. For $s = \pm 1$ and $n \ge 1$ use the principal-power functions

$$\widetilde{h}_n(z) = e^{-n} z^{1-4isn/\pi}(1 + z)^{-3}, \qquad \widetilde{h}_n(0) = 0. \tag{12}$$

Their suprema on the positive and imaginary axes are at most e^{-n} and e^n, respectively. Each is the Laplace transform of an L^1 function. Here is the factorization that verifies this point, including the boundary of the half-plane. For $\nu = 1 - 4isn/\pi$,

$$z^\nu(1 + z)^{-3} = \big(1 - (1 + z)^{-1}\big)^\nu (1 + z)^{-(3-\nu)}.$$

The second factor is the transform of $a^{2-\nu}e^{-a}/\Gamma(3 - \nu)$, an integrable function. The binomial coefficients of the first factor are absolutely summable, since $|\binom{\nu}{k}| = O(k^{-2})$. It is therefore the transform of the finite complex measure

$$\delta_0 + \sum_{k \ge 1} (-1)^k \binom{\nu}{k} \frac{a^{k-1}e^{-a}}{(k - 1)!} \, \mathrm{d}a.$$

Convolution gives $h_n \in L^1$. Absolute convergence and $\operatorname{Re}\nu = 1$ also give the stated boundary value at $z = 0$.

Fix $\delta > 0$ and the sector $S_\delta = \{z \ne 0 : \operatorname{Re} z \ge 0, \ s \arg z \ge \pi/4 + \delta\}$. On this sector,

$$|\widetilde{h}_n(z)|^2 \ge e^{8\delta n/\pi}|z|^2|1 + z|^{-6}.$$

Substitution in (11) gives

$$e^{8\delta n/\pi} \int_{S_\delta} e^{-2DE}|z|^2|1 + z|^{-6} \, \mathrm{d}\varrho \le \|A\|_2^2.$$

The integrand is positive on S_δ, so letting $n \to \infty$ shows $\varrho(S_\delta) = 0$. Taking both signs and rational $\delta > 0$ gives $|\arg z| \le \pi/4$ on its support, or $E \ge |p_i|$.

Finite linear combinations of characters with compact positive-side tests lie in some such B and have dense quotient classes in $\mathcal{H}_j$. The spectral projection of $\{E < |p_i|\}$ vanishes on this dense set and hence vanishes identically. □

Applying this result to the nine-mirror laws gives $H_j \ge |\mathsf{P}_i|$ for every ordered pair $i \ne j$. This coordinate cone identifies the invariant projections used in clustering.

4. Invariant observables and clustering

Let P_j be the orthogonal projection in $L^2(\lambda)$ onto the observables fixed by all $U(te_j)$, and let P project onto those fixed by all translations. These are the corresponding conditional expectations by lemma A.2. The averages $T^{-1}\int_0^T U(te_j)\,\mathrm{d}t$ converge strongly to P_j. The projections commute with one another and with every translation.

An observable fixed by translations along e_j can be represented in either half-space bounded by any plane $x_j = c$. Indeed, approximate it in L^2 by finite sums of characters with compact tests, and translate each approximant far enough into the chosen half-space. Translation fixes the limiting observable and preserves the approximation error. The half-space L^2 subspace is closed.

The linear geometric reflection $\mathcal{R}_j$ restricts to a positive self-adjoint involution on this invariant subspace. Positivity follows from its inclusion in the positive-side space and reflection positivity. Its spectrum is therefore $\{1\}$, so

$$\mathcal{R}_j A = A \qquad (A \in \operatorname{ran} P_j). \tag{13}$$

Thus the quotient map is an isometry there. Its range is precisely $\ker H_j$. To identify it, average

$$\langle [A]_j, e^{-tH_j}[B]_j\rangle = \langle \mathcal{R}_j A, U(te_j)B\rangle_{L^2(\lambda)}$$

over $0 \le t \le T$. Spectral dominated convergence on the left and lemma A.2 on the right give

$$P_0[B]_j = [P_j B]_j, \qquad P_0 = \mathbf{1}_{\{0\}}(H_j). \tag{14}$$

The image of $\operatorname{ran} P_j$ is closed because the map is isometric, and contains the dense subspace $P_0[\mathcal{E}_j^+]_j$ of $\ker H_j$. This proves the asserted identification.

For a law invariant and reflection positive at the nine mirrors, the cone forces the joint spectral measure at zero energy to be concentrated at zero momentum. Hence $V_j(y)$ fixes $\ker H_j$. If $A \in \operatorname{ran} P_j$ and $y \perp e_j$, then

$$[U(y)A - A]_j = 0.$$

The difference is again in $\operatorname{ran} P_j$, so the isometry gives $U(y)A = A$. Consequently

$$P_1 = P_2 = P_3 = P. \tag{15}$$

4.1. Correlations in all directions.

Proposition 4.1. Suppose λ is translation invariant and invariant and reflection positive at the nine mirrors. For all $A, B \in L^2(\lambda)$,

$$\langle A, U(a)B\rangle \longrightarrow \langle A, PB\rangle \quad \text{as } |a| \longrightarrow \infty. \tag{16}$$

If λ is translation ergodic, the limit is $\overline{\mathbb{E}_\lambda A}\,\mathbb{E}_\lambda B$ and $\ker H_j = \mathbb{C}\Omega_j$ for every j. ■

Proof. First fix j. For $A = \mathcal{R}_j A'$ and $A', B \in \mathcal{E}_j^+$, (14) gives, for $t \ge 0$ and $y \perp e_j$,

$$\begin{aligned}|\langle A, U(te_j + y)(1 - P_j)B\rangle| &= |\langle [A']_j, V_j(y)e^{-tH_j}(1 - P_0)[B]_j\rangle| \\ &\le \|[A']_j\|\,\|e^{-tH_j}(1 - P_0)[B]_j\|.\end{aligned}$$

The last norm tends to zero by spectral dominated convergence; the bound is uniform in y. Approximate general A, B by local character sums. Translate their supports into opposite half-spaces, at a cost of replacing t by $t - 2r$ for one fixed r. The correlation error is bounded uniformly in t, y by $\|A - A_1\|_2\|B\|_2 + \|A_1\|_2\|B - B_1\|_2$. The adjoint identity gives the same conclusion as $t \to -\infty$. Thus

$$\lim_{|t|\to\infty} \sup_{y \perp e_j} |\langle A, U(te_j + y)(1 - P_j)B\rangle| = 0.$$

By (15), $P_j = P$ for all j. For any sequence $|a| \to \infty$, a subsequence has one fixed coordinate tending in modulus to infinity. The displayed uniform bound applies with that coordinate, while $U(a)PB = PB$. Every subsequence has such a further subsequence, proving (16). In an ergodic law P is the projection onto constants. The isometry onto $\ker H_j$ then gives its one-dimensionality. □

5. Ergodic laws and inherited reflection forms

We recover the reflection properties of conditional laws by testing the parent reflection forms against invariant weights.

Let $\mathcal{I}$ be the translation-invariant event sigma algebra. The evaluation space is standard Borel by lemma A.1. The regular conditional distribution theorem [Dur19, Theorems 4.1.16–4.1.17] therefore gives conditional laws λ_F given $\mathcal{I}$, with

$$\lambda_F(\Phi) = \mathbb{E}_\lambda[\Phi \mid \mathcal{I}](F), \qquad \lambda = \int \lambda_F \, \mathrm{d}\lambda(F). \tag{17}$$

The first equality is understood almost surely for each integrable Φ (and for nonnegative Φ by monotone convergence). The image measure on laws is the translation-ergodic decomposition. We recall why a single full-measure set of ergodic laws can be used.

For an open set O, $\mathcal{S}(\overline{O}; V)$ denotes the subspace of $\mathcal{S}(\mathbb{R}^3; V)$ consisting of functions supported in $\overline{O}$, with the induced Schwartz topology. Fix a countable real test space $\mathcal{D}_0 \subset C_c^\infty(\mathbb{R}^3; V)$ over $\mathbb{Q}$, stable under rational translations and cubic isometries, with $\mathcal{D}_0 \cap C_c^\infty(O; V)$ dense in $\mathcal{S}(\overline{O}; V)$ for $\mathbb{R}^3$ and the six coordinate half-spaces O. Take the rational span of translated cubic images of countable Schwartz-dense compact-test families in these seven open sets to obtain it. Let $\mathcal{A}_0$ consist of finite sums of $e^{iF(f)}$, $f \in \mathcal{D}_0$, with coefficients in $\mathbb{Q} + i\mathbb{Q}$. This same countable family is dense in L^2 for every law on the evaluation sigma algebra and determines such laws. For density, condition an observable orthogonal to its span on finitely many evaluations. Rational combinations of the tests show that the Fourier transform of the resulting finite complex measure vanishes on rational frequencies. Continuity and Fourier uniqueness make it zero. Conditional expectations on the increasing finite evaluation sigma algebras converge to the observable, proving density. The same finite-dimensional Fourier argument proves determination of laws. For the elements A_p, choose cube-average times $T_k \to \infty$ so that

$$\|M_{T_k} A_p - P A_p\|_2^2 \le 2^{-k} \quad (p \le k), \qquad M_T = T^{-3} \int_{[0,T]^3} U(a) \, \mathrm{d}a.$$

Equation (17) and summability give convergence in $L^2(\lambda_F)$ for every p on one full-measure set of F. Moreover PA_p has zero conditional variance, and thus equals the constant $\lambda_F(A_p)$ in that space. The conditional laws are translation invariant: test the parent invariance against $\mathcal{I}$-measurable bounded weights, first for rational translations and the countable characters, then use continuity and Fourier uniqueness. Lemma A.2 for each such law now shows that its invariant projection is the constant projection, first on the dense algebra and then on all $L^2(\lambda_F)$. Thus these conditional laws are ergodic.

To prove uniqueness, the summable parent errors above give $M_{T_k} A_p \to P A_p$ almost surely, simultaneously for all p. Let $\lambda = \int \eta \, \mathrm{d}\pi(\eta)$ be any decomposition into translation-invariant ergodic laws. This parent full-measure event has full η-measure for π-almost every η. On the other hand, lemma A.2 gives $M_{T_k} A_p \to \eta(A_p)$ in $L^2(\eta)$. Uniqueness of the limit in probability therefore gives $PA_p(F) = \eta(A_p)$ for η-almost every F, simultaneously for all p. The conditional-law identity in (17) holds on another parent full-measure event for this same countable family. Thus $\lambda_F(A_p) = \eta(A_p)$ there for every p. Since $\mathcal{A}_0$ determines laws, $\lambda_F = \eta$. Taking the distribution of λ_F under the mixture shows that its image measure is π, proving uniqueness.

Fix an axis mirror. A bounded invariant weight h satisfies $\mathcal{R}_j h = h$ by (13) and has a representative in either half-space. If $h \ge 0$ and $A \in \mathcal{E}_j^+$, reflection positivity applied to $\sqrt{h}A$ gives

$$\mathbb{E}_\lambda[h(\Theta_j A)A] \ge 0. \tag{18}$$

Reflection invariance similarly gives $\mathbb{E}_\lambda[h\mathcal{R}_j A] = \mathbb{E}_\lambda[hA]$. Taking all bounded $\mathcal{I}$-measurable weights in these identities proves that the corresponding conditional reflection form is nonnegative and the conditional character expectations are reflection invariant. Countably many positive-side character

sums suffice; density extends the statements to the full positive-side algebra in each conditional law. Translation invariance moves the mirror to every real height.

This argument also transfers any further fixed mirror with unit normal n. Choose $Q \in O(3)$ with $Qe_1 = n$ and apply the axis argument to the transported law $(Q^{-1})_*\lambda$. The transported law is reflection positive at $x_1 = 0$; it need not equal λ. The map Q_* sends translation-ergodic laws to translation-ergodic laws and transports their reflection forms. By uniqueness of the ergodic decomposition, its image decomposition is that of λ. Applying this to the six diagonal normals, and intersecting the resulting nine full-measure sets, transfers all required mirrors.

The nine reflections generate the cubic group, so these components are cubically invariant. Proposition 4.1 applies to each component and gives clustering and $\ker H_j = \mathbb{C}\Omega_j$. Any specified measurable set of full parent measure has full measure in almost every component by (17). This proves all assertions of theorem A except the optional Gaussian bound.

6. Gaussian bounds in the components

Suppose now that the parent law satisfies (5) with the constant $C \geq 0$ stated there. To transfer this Gaussian domination with the same constant, fix real $f \in \mathcal{S}$ and put $G = e^{F(f)}$, $c = e^{C\|f\|_2^2/2}$, $\phi = PG$. All moments of G are finite. For each integer $n \geq 2$, Tonelli's theorem and the parent bound give

$$\mathbb{E}_\lambda[(M_T G)^n] \leq T^{-3n} \int_{([0,T]^3)^n} \exp\left(\frac{Cn}{2}\|f\|_2^2 + \frac{C}{2}\sum_{p\neq q}\langle f, \tau_{a_p - a_q} f\rangle\right) \mathrm{d}a. \tag{19}$$

The overlaps tend to zero as $|a_p - a_q| \to \infty$ and are bounded by $\|f\|_2^2$. Fix $\epsilon > 0$. Outside the set where one pair is within distance δ, the right side is at most $c^n e^{Cn(n-1)\epsilon/2}$ once δ is sufficiently large. The exceptional set has relative volume at most $\binom{n}{2}(2\delta)^3 T^{-3}$ and its integrand is at most $e^{Cn^2\|f\|_2^2/2}$. Since $M_T G \to \phi$ in L^2, and the averages and ϕ are uniformly bounded in L^{2n}, interpolation gives convergence in L^n. Sending $T \to \infty$ and then $\epsilon \to 0$ in (19) yields $\|\phi\|_n \leq c$ for every n, hence $\phi \leq c$ almost surely. By (17),

$$\lambda_F(e^{F(f)}) \leq e^{C\|f\|_2^2/2} \quad \text{almost surely.} \tag{20}$$

Choose a countable Schwartz-dense real test family and intersect these full-measure sets. For any other test, pointwise convergence of the evaluations under each conditional law and Fatou's lemma extend (20) to it. Thus all real tests obey the bound on one full-measure set of component laws.

This proves the Gaussian assertion of theorem A. For a two-form field, closedness also passes to the components whenever it holds in the parent law: the event $dF = 0$ is measurable by testing it on a countable dense family of three-form tests.

Appendix A. Measurability and translation averages

We give the measure-theoretic details used in the component argument. They require no moment bound.

Lemma A.1. The space $\mathcal{X} = \mathcal{S}'(\mathbb{R}^3; V)$, with its evaluation sigma-algebra, is standard Borel. Translation is a jointly measurable action on this space. ∎

Proof. Choose a countable rational vector space $\mathcal{D}_0$ dense in $\mathcal{S}(\mathbb{R}^3; V)$, and increasing norms q_n defining the Schwartz topology. For $n \geq 1$, put

$$X_n = \{F : |F(f)| \leq n q_n(f) \text{ for every } f \in \mathcal{S}\}.$$

Continuity of each distribution gives $\mathcal{X} = \bigcup_n X_n$. The evaluation map $\iota(F) = (F(f))_{f\in\mathcal{D}_0}$ is injective into the Polish space $\mathbb{R}^{\mathcal{D}_0}$. Its image on X_n is exactly the set of sequences satisfying

$$z_{sf+tg} = s z_f + t z_g, \qquad |z_f| \leq n q_n(f) \quad (f, g \in \mathcal{D}_0,\ s, t \in \mathbb{Q}).$$

These are closed conditions on countably many coordinates. Conversely, the bound extends such a rational linear functional continuously and uniquely to the Schwartz space; density and continuity make the extension real linear. Thus $\iota(\mathcal{X})$ is a countable union of closed sets. Every evaluation is measurable with respect to the $\mathcal{D}_0$ coordinates by approximation and samplewise continuity. Hence ι is a measurable isomorphism onto a Borel subset of a Polish space, which is standard Borel [Dur19, Theorem 2.1.22].

For fixed f, the function $(a, F) \mapsto F(\tau_{-a} \cdot f)$ is continuous in a and measurable in F. Replace a by its coordinatewise lower dyadic approximation $2^{-n}\lfloor 2^n a \rfloor$. The resulting functions are measurable on the product space and converge pointwise to the original one. Since evaluations generate the sigma-algebra, the translation action is jointly measurable. □

Lemma A.2. Let λ be translation invariant. If $J \subset \{1, 2, 3\}$ is nonempty, the averages

$$M_T^J A = T^{-|J|} \int_{[0,T]^J} U(a)A \, \mathrm{d}a$$

converge in $L^2(\lambda)$ to the orthogonal projection P_J onto the observables fixed by the translations in the coordinate subspace spanned by $\{e_j : j \in J\}$. This projection is conditional expectation onto the corresponding invariant sigma-algebra. All M_T^J are contractions on $L^p(\lambda)$, $1 \le p < \infty$. ■

Proof. Joint measurability and Fubini give the averages, and Jensen's inequality gives the L^p contraction. The representation U is strongly continuous by bounded convergence on local characters and their density in $L^2(\lambda)$. Let $\mathcal{K}$ be the closed span of $B - U(b)B$, where b is in the coordinate subspace. Its orthogonal complement is the invariant subspace. The averages fix that subspace. The symmetric-difference bound for a cube and its translate gives

$$\|M_T^J(B - U(b)B)\|_2 \le \frac{2|b|_1}{T}\|B\|_2.$$

Thus the averages tend to zero on a dense subspace of $\mathcal{K}$, and their uniform norm bound extends this convergence to all of $\mathcal{K}$. This proves the asserted orthogonal-projection limit.

The invariant subspace equals the L^2 space of the invariant sigma-algebra: indicators of invariant events are fixed, and the inverse images of Borel sets under a fixed observable are invariant modulo null sets. Therefore its orthogonal projection is the stated conditional expectation. The coordinate projections commute, since their averaging operators commute. □

References

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